WASP-50b

R-0861

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-50_180105` · created 2026-07-28T22:13:57+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458123.694 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1503 +/- 0.009
Transit depth (Rp/Rs)^2	2.26 +/- 0.27 [%]
Orbital Inclination (inc)	84.62 +/- 0.43
Airmass coefficient 1 (a1)	1.1836 +/- 0.0092
Airmass coefficient 2 (a2)	-0.1133 +/- 0.0032
Scatter in the residuals of the lightcurve fit is	0.77 %
Best Comparison Star	None
Optimal Aperture	4.0
Optimal Annulus	11.89
Transit Duration (day)	0.0753 +/- 0.0037

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1503 ± 0.009 vs 0.1404 (deviation 0.92σ)
Duration fit vs published	0.0753 d vs 0.0754 d (deviation 0.03σ)
Mid-time O–C	-0.8 min
Depth SNR	13.9
Residual scatter	0.77 %

Light curve

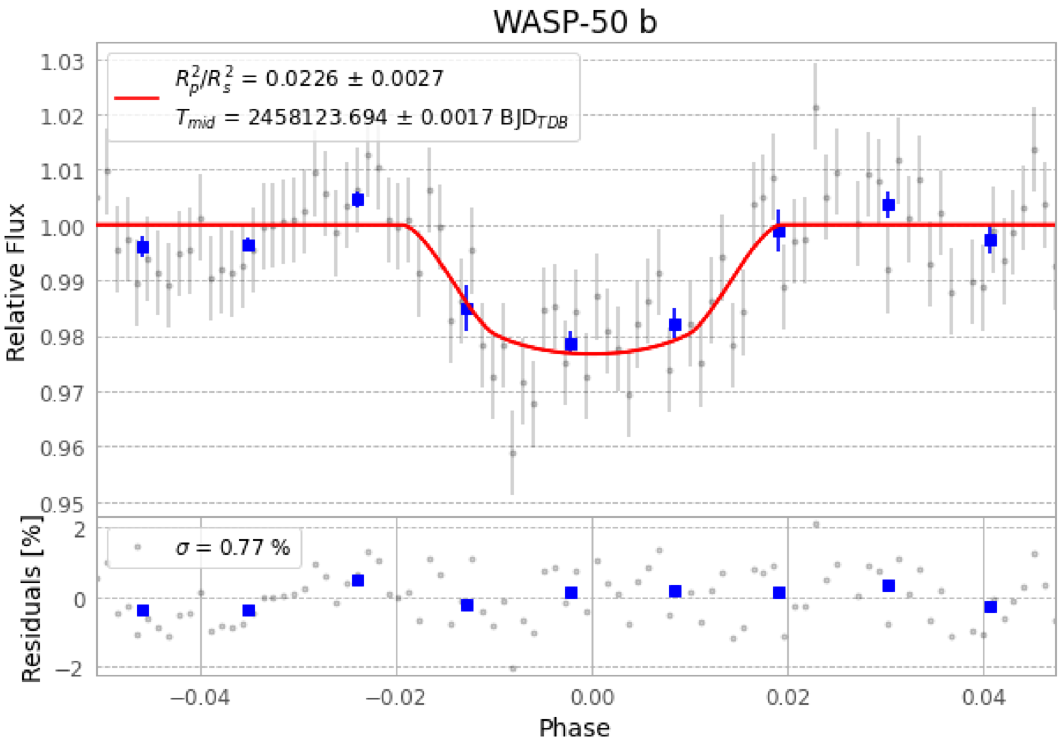

Figure 1 — FinalLightCurve_WASP-50 b_2018-01-04.png (SHA-256 be96fe838bdd11fe…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [281, 287], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 1abad061f6c1b0cb…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

94 FITS frames (62 MB), `WASP-50180105021212.FITS` → `WASP-50180105065112.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-180105.log (fb098b7721ef…), FinalLightCurve_WASP-50 b_2018-01-04.png (be96fe838bdd…), FinalLightCurve_WASP-50 b_2018-01-04.csv (d4599e7dd297…)

Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)